

Decision Trees

Project: Responsible Plant Classifier

1. Decision Trees

Teach the learning cycle

Primary target: Fit a root question and leaf predictions from training examples

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/14, ai-2/15, ai-4/11. Reuse a familiar case for

recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Baselines First

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a majority label baseline using training labels only

Reference example, not the only acceptable wording: Training has four clear and two blocked, so the constant is clear. A tie chooses clear under the declared classroom rule. Validation labels score the stored prediction; using their majority to choose it would change the fitting procedure.

Leakage

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended prediction claim

Reference example, not the only acceptable wording: Training fits the model; validation compares development choices; test evaluates after choices are fixed. Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Rule: if spots $\geq$ 4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that training found the threshold?

Concept to explain

A decision tree partitions feature space using ordered questions; each leaf produces a prediction.

Term · Meaning

Term: split

Meaning: question dividing examples

Term: threshold

Meaning: numeric boundary used in a split

Term: leaf node

Meaning: terminal prediction

Term: pruning

Meaning: remove unnecessary branches

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Students fit a depth-one decision tree by learning branch majorities and selecting the question with the fewest total training mistakes. This transparent classroom algorithm is not a claim to reproduce CART or a library default. Explain tuple positions, list comprehension and the leaves dictionary; offer the full paper route for students who need it.

Readiness answer

The cutoff was selected using training labels. Matching those examples does not establish performance on new examples; validation can expose mistakes and support a baseline comparison.

Printed lesson and workbook

Six training row cards and four validation cards

Two branch areas labelled 0 and 1

Optional Python 3 editor with no packages required

End goal for the pictorial case

By the end, I can explain decision trees using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Trace, draw, and prune a tiny tree while preserving test behavior.

MODEL IDEA A decision tree partitions feature space using ordered questions; each leaf produces a prediction.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Root asks whether $\text{spot_count} \geq 3$. Yes branch predicts spotted; no branch asks whether $\text{yellow_ratio} \geq 0.25$.

Draw a fresh data $\rightarrow$ model $\rightarrow$ prediction $\rightarrow$ evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Root asks whether $\text{spot_count} \geq 3$.

STEP 2 Yes branch predicts spotted; no branch asks whether $\text{yellow_ratio} \geq 0.25$.

STEP 3 Trace leaf A: $\text{spot_count}=1$, $\text{yellow_ratio}=0.10 \rightarrow$ healthy.

STEP 4 A branch fitting one training oddity hurts test results, so prune and retest.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

A supplied tree predicts spotted for $\text{spot_count} \geq 3$, otherwise healthy.

Trace 2,3,4 and state boundary.

Exact explanation to model

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned threshold.

Explain the added worked example

A decision tree asks a feature question and follows a branch to a prediction. A tree with one question has depth one and is sometimes called a stump. Unlike Week 8, where the two output labels were fixed in

advance, this week we fit both the question and the label stored at each leaf.

Our six fictional training rows have two binary sensor flags available before prediction: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate. Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Try near first and tall second. For each question, group the training rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts. Count all training mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom fitting procedure.

Feature 0 is near; feature 1 is tall; row[2] is the label. Output is 0 1, then 1 2, then 0 clear blocked. Near makes one training mistake: near=0 contains two clear labels and one blocked, so that leaf predicts clear. Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both branches of each candidate. An empty branch would need a declared fallback; this small example does not implement one. The score is total training mistakes, not an average giving small and large branches equal weight.

This is an explicitly limited depth-one learner. General tree algorithms can repeat split selection within branches and often use impurity criteria such as Gini. Our code does not implement that full procedure. Keeping the scope small lets you inspect every decision.

Key terms

Term · Meaning

Term: Root

Meaning: The first question asked by a tree

Term: Branch

Meaning: A path selected by the answer to a question

Term: Leaf

Meaning: An endpoint that stores a prediction

Term: Depth

Meaning: The largest number of question steps from root to leaf

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Two paths through the learned tree

Root question · Branch answer · Learned leaf

Root question: Is near equal to 1

Branch answer: No near is 0

Learned leaf: Predict clear

Root question: Is near equal to 1

Branch answer: Yes near is 1

Learned leaf: Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: predictions are blocked, clear, clear, blocked. Two of four match. The training-majority baseline predicts blocked, since training has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Keep the fitted root and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct training predictions out of six is not the validation result.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Decision Trees

Trace a small fitted tree, including its exact boundary.

The facts we will use

Toy stored tree: if visible_spots ≥ 2 predict spotted; otherwise healthy. Missing visible_spots requires review. Inputs L=1, M=2, N=None.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Decision Trees

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read first branch

L: $1 \geq 2$ is false, so predict healthy.



2. Read boundary

M: $2 \geq 2$ is true, so predict spotted.



3. Stop missing path

N lacks a count, so use review rather than inventing zero.

CHANGE ONE THING

The stored rule changes to $\text{visible_spots} > 2$.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read first branch

What happens and why: L: $1 \geq 2$ is false, so predict healthy.

Step: 2. Read boundary

What happens and why: M: $2 \geq 2$ is true, so predict spotted.

Step: 3. Stop missing path

What happens and why: N lacks a count, so use review rather than inventing zero.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Applying a stored tree is inference. Selecting a threshold from labelled training data is a separate fitting step.

Change one thing in the pictured case

Change: The stored rule changes to `visible_spots > 2`.

Prediction: M is now predicted healthy.

Reason: Changing `>=` to `>` moves the exact value 2 to the other branch.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A supplied tree predicts spotted for `spot_count >= 3`, otherwise healthy.

Write the question to explain: Trace 2,3,4 and state boundary.

Reveal after predictions: Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned threshold.

Keep for an independent attempt: A tree predicts fern when `leaf_width_cm <= 3` and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: tree predictions blocked,clear,clear,blocked miss rows 3 and 4. Baseline blocked misses rows 2 and 4. Both get 2/4 or 50 percent. Tall cannot affect prediction because the stored root uses near only. Exit: the near=1 leaf contains three blocked training examples, but the whole tree gets 2/4 on this tiny validation set and misses one near=1 validation case. This does not establish reliability on future trips.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve training search versus validation. Introduce root, branch, leaf and depth with one question.

Stage: 7 to 18 minutes

Teacher action: Sort the six cards by near. Count labels within each branch and assign its majority without using validation.

Stage: 18 to 32 minutes

Teacher action: Repeat for tall and complete W1 and W2. Add branch mistakes to score the full training set.

Stage: 32 to 45 minutes

Teacher action: Trace code and the diagram. Predict all four validation rows, complete the guided question and compare the baseline.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve why a leaf label is learned rather than dictated by the feature name.

Stage: 8 to 20 minutes

Teacher action: Complete W3 and discuss depth limits, tied questions and the assumptions required by the short code.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before hints. Require branch counts, both candidate scores, learned leaves and validation comparison.

Stage: 35 to 45 minutes

Teacher action: Use W5 and the exit claim. Give the fresh retry after feedback and record support.

Reduce help deliberately

Completion task: For each near branch and each tall branch, count blocked and clear labels and choose a leaf. Give the total errors for each candidate question.

Teacher check: Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear. Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The branch label is learned from its training labels; W4 deliberately reverses that relationship.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which input distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

M with exactly 2 spots. Values 1 and missing keep their earlier declared paths; 2 is the boundary probe. Changed-case check: M is now predicted healthy. Changing $\geq$ to $>$ moves the exact value 2 to the other branch.

Changed-case reasoning: M is now predicted healthy. Changing $\geq$ to $>$ moves the exact value 2 to the other branch.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Human Tree Builder: choose splits, trace six examples, compare train/test errors, then prune one weak branch.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A deeper tree must be better because it asks more detailed questions.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A near branch must predict blocked: The branch label is learned from its training labels; W4 deliberately reverses that relationship.

Every feature is used during prediction: Only questions in the stored tree are followed. Candidate questions rejected during fitting do not affect this tree's output.

A deeper tree must work better: More questions can fit training detail. Future performance needs separate evaluation; depth choices must be documented.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed

incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Healthy, spotted, spotted. Three is included. This supplied rule does not show an optimal learned threshold.

Independent question

A tree predicts fern when $\text{leaf_width_cm} \leq 3$ and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

Expected independent answer

Widths 2 and 3 predict fern; 4 predicts grass. Width 3 checks the inclusive boundary.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “An inclusive threshold excludes an example exactly equal to it.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Outputs are healthy, spotted, spotted, and review or an explicitly defined missing-value path. Tracing applies a rule. To claim training, show how candidate thresholds were evaluated on training examples and how the chosen threshold was selected.

Explanations for the additional workbook examples

W1 Name the parts and decisions

Identify the root, its two branches, leaf predictions and depth of the worked fitted tree. Which parts were selected using training labels?

The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one. Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

W2 Verify the candidate scores

For each near branch and each tall branch, count blocked and clear labels and choose a leaf. Give the total errors for each candidate question.

Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear. Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

W3 Separate fitting from prediction

A learner sees a blocked validation case with near=0 and changes that leaf to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

The leaf was changed using validation evidence, so it is no longer the original training-fitted tree. Score the original stored leaves on all the same validation rows and record any later development change separately. Preserve a separate final test for final evaluation.

W4 Fit a new tree independently

New training rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree. Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree predictions and compare accuracy with the training-majority baseline.

Near=0 predicts blocked from 2 blocked and 1 clear; near=1 predicts clear from 3 clear. Near has 1 error. Both tall groups have 2 clear and 1 blocked, so both leaves predict clear and total errors are 2. Choose near with reversed leaf labels compared with the worked example. Validation predictions clear,blocked,blocked,clear match rows 1,2: 2/4. Baseline clear matches rows 1,3: 2/4. Both are 50 percent but errors differ.

W5 Audit a tie and a claim

If near and tall make the same training error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

It retains near because near is tried first and strict < replaces only for fewer errors. This is a declared tie convention, not evidence of better future performance. Separate evaluation and relevant examples are still needed.

Assess independent transfer

Selected case: A stored fitted rule says spots ≥ 5 predicts spotted; otherwise healthy. Trace 4, 5, 6 and a missing count. Policy: missing count requires review. Can test labels be used to choose this cutoff?

Selected case answer

4 predicts healthy; 5 and 6 predict spotted; missing requires review. Applying the stored cutoff is inference, not fitting. Choosing it using test labels contaminates the final evaluation; use designated development data.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Draw the current Plant Classifier tree and annotate feature, threshold, prediction, and test evidence at

each leaf.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.
Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 12: Nearest Neighbors. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use physical sorting before code. Explain `row[feature]` as selecting a flag and `row[2]` as reading its label. Supply empty branch-count tables. Extension: explain why an empty branch needs a fallback, and why this code would silently return clear if given an empty label list.

Require W4 near error 1, tall error 2, near=0 blocked and near=1 clear, with both validation scores 2/4. Fresh retry: training (0,0,clear),(0,1,blocked),(1,0,clear),(1,1,blocked). Near branches each tie and predict clear, giving 2 errors. Tall=0 predicts clear and tall=1 blocked with 0 errors; choose tall. Validation (0,1,blocked),(1,0,clear),(1,1,clear) gives blocked,clear,blocked: 2/3. The tied training baseline predicts clear and also gets 2/3, with different missed rows. Require the learned feature and both tie policies.

Week 10 introduces nearest neighbours and distances. Carry forward the training/validation split and baseline. Teacher reference: <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html> (reviewed 2026-09-27); its library criteria differ from our stated classroom mistake-count procedure.

Plan the delayed return

Suggested return points in this level: ai-4/12, ai-4/14, ai-4/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record - Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Prepare, teach and assess

Teaching target: Fit a root question and leaf predictions from training examples

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For each near branch and each tall branch, count blocked and clear labels and choose a leaf. Give the total errors for each candidate question.

Reasoned practice answer: Near=0 has 1 blocked and 2 clear, predicts clear with 1 error; near=1 has 3 blocked and 0 clear, predicts blocked with 0 errors. Tall=0 has 2 blocked and 1 clear; tall=1 also has 2 blocked and 1 clear. Both tall leaves predict blocked with 1 error each. Totals are near 1 and tall 2.

Check every learner before discussion: Use the worked case for Fitting a decision tree. Proposed explanation: “A near branch must predict blocked”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The branch label is learned from its training labels; W4 deliberately reverses that relationship.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The stored rule changes to `visible_spots>2`. Require a prediction and a reason before comparing. Expected result: M is now predicted healthy. Reason: Changing `>=` to `>` moves the exact value 2 to the other branch.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.